

New Product Preannouncements (NPPAs) and the Social Amplification of Risk: Evidence from French COVID-19 Tweets

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Research Object: NPPAs as Strategic Signals

- **NPPA** = deliberate pre-launch communication to shape expectations and market behavior (Eliashberg & Robertson, 1988; Robertson et al., 1995; Su & Rao, 2010).
- **Signaling theory**: firms use signals to reduce information asymmetry in uncertain markets (Akerlof, 1970; Spence, 1973).
- In pharmaceuticals, signals target investors, regulators, competitors, health professionals, and the public.

Strategic Benefits vs. Reputational Risk

- **Benefits:** reassure stakeholders, deter competitors, and frame innovation before launch.
- **Risks:** once public, messages can be reframed by audiences, shifting attention toward safety concerns and controversies.
- Early communication is a strategic trade-off rather than a purely positive signal.

Social Amplification of Risk (SARF)

- SARF explains how risk perceptions are transformed by social processes, not only by technical facts (Kasperson et al., 1988).
- Digital platforms act as **amplification stations**, which can amplify or attenuate perceived risk (Kasperson et al., 2022).
- SARF frames amplification/attenuation as a two-stage process: initial signal transfer and subsequent societal responses (behavioral, economic, symbolic) (Hopfer et al., 2025).
- Early amplification is shaped by information volume, ambiguity/dispute, dramatization, and symbolic cues (Hopfer et al., 2025).
- This creates a tension between **controlled signals** and **emergent public narratives**.

Why COVID-19 Is a Stress Test

During the COVID-19 pandemic, the global demand for a vaccine created immense public and market pressure.

- **Massive market potential:** global market in need of billions of doses => **high R&D costs.**
- **Accelerated regulatory processes:** Agencies like the **EMA** fast-tracked approval procedures to respond to the crisis.
- **Intense competition:** A race to develop the first vaccine to secure **first mover advantage**

Research Questions

Research question	What is measured?	Operationalization (data + methods)
RQ1 / H1 (structure): Who are the amplification stations / opinion leaders?	<i>Who amplifies whom</i> in the COVID/vaccine issue arena	Retweet network (directed, weighted): nodes = accounts; edge $u \rightarrow v$ if u retweets v (weight = count).
RQ2 / H2 (social reaction): How do Twitter users react to NPPAs?	Cognitive / emotional responses around announcements	Sentiment/emotion analysis in NPPA event windows
RQ3 / H3 (risk amplification): How is risk-related content amplified?	Salience of side effects and brand associations vs clinical signals	Adverse-event extraction; side-effect categories; brand-specific associations; comparison with official signals (SARF gap).

Methodology Overview

- **Data:** Full corpus of 50.4M French tweets (2020-2021; Banda et al., 2021). Retweet network uses the full corpus; content analyses use ~150,000 unique vaccine-brand tweets after filtering and removing RTs/duplicates.
- **Side Effect Extraction (NER):** Used **GLiNER**, a zero-shot NER model (Zaratiana et al., 2023), fine-tuned on a small annotated set (100 tweets, F1=0.81), to identify side-effect mentions without large specific training data. Side effects were standardized into categories.
- **Sentiment Analysis:** Used **XLM-RoBERTa** (Barbieri et al., 2021) (fine-tuned for multilingual COVID-19 tweets) to classify tweets as positive, neutral, or negative. Analyzed sentiment shifts “Before”, “On”, and “After” key announcement dates.

Dataset Overview (Full Corpus)

metric	value
Tweets (N)	50,376,815
Unique users	3,022,144
Date range	2020-01-29 to 2021-11-30
Verified share (tweets)	1.6%
Retweets (share)	72.3%
Originals (share)	11.8%
Replies (share)	8.1%
Quotes (share)	7.9%
Median retweets	37
Median likes	77
Mean retweets (p99.9 cap)	1107.1
Mean likes (p99.9 cap)	3809.1
Avg words	25.8
Avg chars	157.2
Avg hashtags	0.79
Avg mentions	0.43
Avg URLs	0.77

Methodology - Retweet Network Construction

- **Graph:** Directed, weighted; edge $u \rightarrow v$ if u retweets v (weight = retweet count).
- **Scale:** 502,565 accounts; GCC (giant connected component) = 475,924 (94.7%) with 3.4M directed edges.
- **Amplification score:** z-score average of weighted indegree + PageRank + k-core (proxy for amplification stations).
- **Diffusion score:** z-score of weighted outdegree (proxy for active relays).

Methodology - NER with Label Studio

Start Label Studio with:

label-studio start

The screenshot displays the Label Studio web interface. On the left is a sidebar with a menu containing: General, Labeling Interface (highlighted), Annotation, Model, Predictions, Cloud Storage, Webhooks, and Danger Zone. The main area is titled 'Projects / Vaccine TRAIN Annotation Effects / Settings / Labeling Interface'. It features a grid of tool cards for different tasks: Computer Vision, Natural Language Processing (selected), Audio/Speech, Conversational AI, Ranking & Scoring, Structured Data, Parsing, Time Series Analysis, Videos, and Generative AI. Below these is a 'Custom template' link. The right side shows a preview of the selected tool, 'Natural Language Processing', with a text input area containing a paragraph about a black hole. A taxonomy dropdown is open, showing a list of categories: Archaea, Bacteria, Eukarya, Human, Opossum, and Extraterrestrial (selected). The interface also includes a 'Question Answering' section with a text input and a 'Text Classification' section with a sentiment selector (Positive, Negative, Neutral).

Label Studio interface

Label Studio

General

Labeling Interface

Annotation

Model

Predictions

Cloud Storage

Webhooks

Danger Zone

Projects / Vaccine TRAIN Annotation Efforts / Settings / Labeling Interface

Computer Vision

Natural Language Processing

Audio/Speech Processing

Conversational AI

Ranking & Scoring

Structured Data Parsing

Time Series Analysis

Videos

Generative AI

Custom template



Semantic Segmentation with Polygons

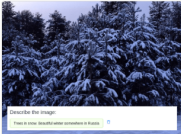


Image Captioning



Invoice Annotation



Semantic Segmentation with Masks



Image Classification



Panda Classification



Object Detection with Bounding Boxes



Inventory Tracking



Q&A Annotation



Keypoint Labeling



Multi-page document annotation

Finetuning - Labelling data

  Projects / Vaccine TRAIN Annotation Effets / Labeling Settings CA

☐



text





☐	1	Génocide en cour : Rapport du gouvernement britannique. Effets	
☐	1	La Norvège s'inquiète d'hémorragies cutanées chez des jeunes ayant reçu	
☐	1	Bon le #vaccin de Pfizer/Biontech est mise en cause dans au moins 13	
☐	1	Un garçon de 13 ans meurt pendant son sommeil le lendemain de sa	
☐	1	Jeune homme de 19 Après sa 2ème dose de Pfizer, mal de tête extrême, nausées,	

#2  + CA caron.olivier.80 #2 8 days ago 

Effet_Secondaire 1

La Norvège s'inquiète d'hémorragies cutanées chez des jeunes ayant reçu du vaccin AstraZeneca ?

Regions

History

Relations

Info

 Manual  By Time 

 1 Effet_Secondaire hémorragies cutanées

Update

Finetuning - Upload annotated data

We used the GliNER Studio Colab notebook from Knowledgator.

File Upload and Save Example

Upload your file here


Déposer le Fichier Ici
- ou -
Cliquer pour Télécharger

Save File

Result

Use via API  · Construit avec Gradio 

Finetuning - Train the model

GLiNER Training Interface

Choose the parent model

urchade/gliner_multi-v2.1

The name of your custom model

Enter the name of your new model

Choose the dataset

Train/Test Split Ratio

0.9

0.1 0.9

Learning Rate

0,000005

0.000001 0.0001

Weight Decay

0,01

0 0.1

Batch Size

8

1 128

Number of Epochs

1

1 10

☐ Compile Model for Faster Training

Start Training

Training Info

Use via API  · Construit avec Gradio 

RQ1 - Results: Retweet Network Topology

metric	value	interpretation
n_nodes	502,565	Total unique accounts (nodes).
n_gcc	475,924	Accounts in the giant connected component (GCC).
gcc_pct	94.70	Share of nodes in the GCC (main connected core).
gcc_edges_directed	3,415,352	Directed ties in the GCC (A->B retweet link).
gcc_edges_undirected	3,385,853	Same ties ignoring direction ({A,B}).
density_directed	0.00001508	Directed density: observed / possible ties (very sparse).
density_undirected	0.00002990	Undirected density: observed / possible links (very sparse).

- The network is **massive** yet **extremely sparse**, with a **dominant giant component**: most accounts are connected through retweet paths.
- This architecture is consistent with **centralized amplification**, where visibility concentrates around a limited set of stations

RQ1 - Results: Amplification Stations (Composite Score, Top 10)

account	amplification_score
Mediavenir	327.766954
f_philippot	106.112639
MediavenirGN	99.256252
FocusFR_	84.167662
LPLdirect	53.739009
franceinfo	51.262677
GaumontRene	49.303869
ConflitsFrance	48.679994
BFMTV	47.629672
ActuFoot_	45.797989

- The composite score captures **robust stations** that remain top-ranked across **volume (attention)** and **structure (centrality/core)**.
- Dominant station types include **media**, **political figures**, and **health professionals** (bio-based inference).

RQ1 - Results: Different Network Roles (Sources vs Relays)

Source-like stations (HITS authorities, Top 10)
(retweeted by strong hubs)

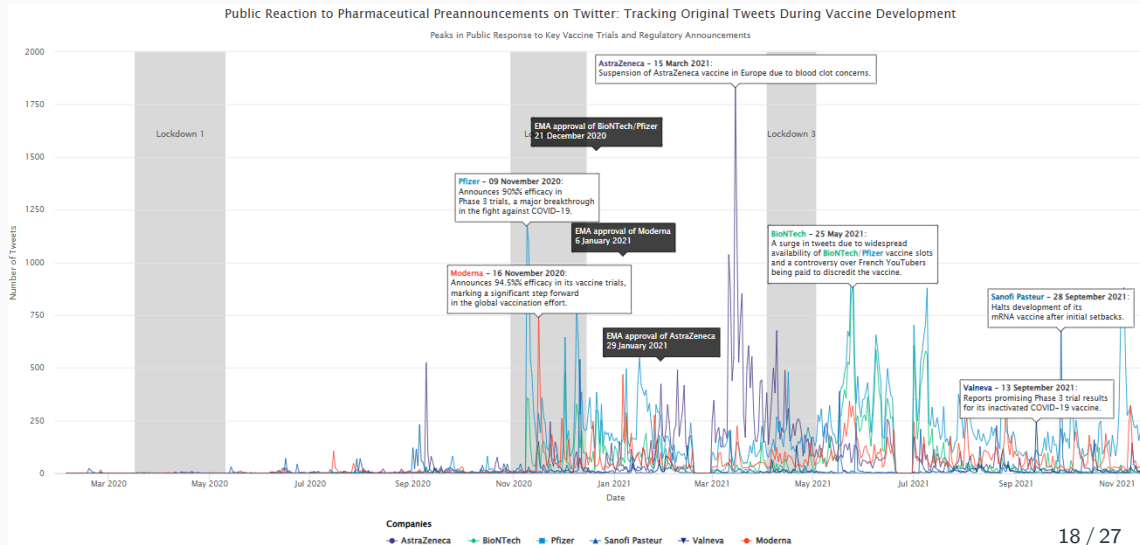
account	authority
Mediavenir	0.951057
ConflitsFrance	0.149024
BFMTV	0.122332
f_philippot	0.091110
le_Parisien	0.070293
franceinfo	0.055892
Laissonslespre1	0.049848
morandiniblog	0.048183
DIVIZIO1	0.046031
LPLdirect	0.045813

High-activity relays (diffusion_score, Top 10)
(propagation channels; heavy retweeters)

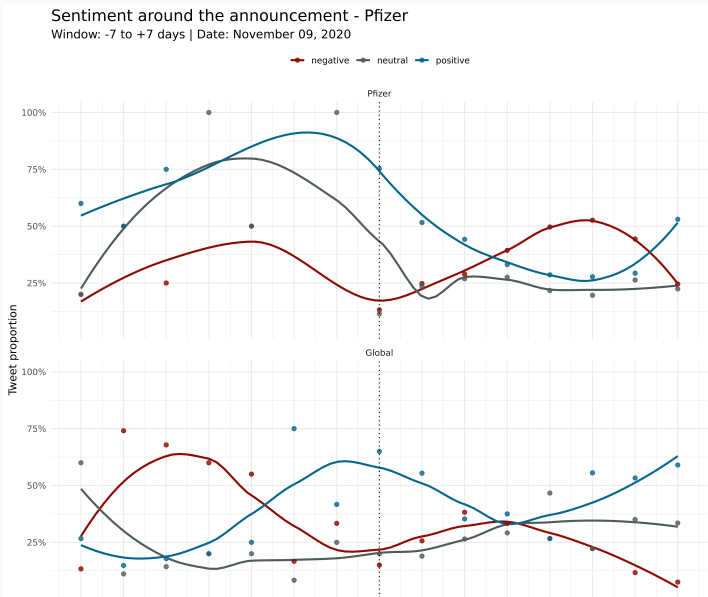
account	diffusion_score
viralvideovlogs	131.296391
julien0686	103.110291
menuet_morgan	90.673560
William85837998	67.635671
vicmahec	67.356033
radiolondres6	64.896651
chriss17031	64.559651
NC4415	64.032640
languillem	61.716662
lagrostA	55.087802

- The network separates **attention sources** (authorities) from **propagation engines** (relays).
- This distinction matters for RQ2/RQ3: sources can shape *what* becomes salient, while relays shape *how far* / *how fast* it travels.

Context Timeline (NPPA Windows)



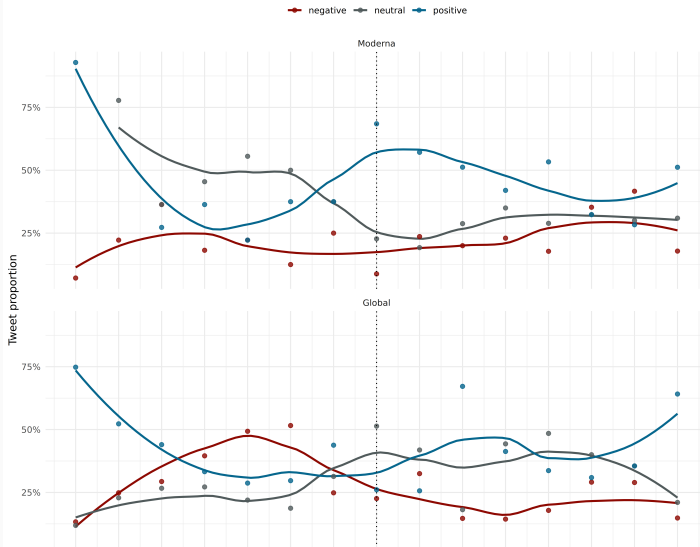
RQ2 - Results: Sentiment Dynamics (Pfizer Efficacy)



RQ2 - Results: Sentiment Dynamics (Moderna Efficacy)

Sentiment around the announcement - Moderna

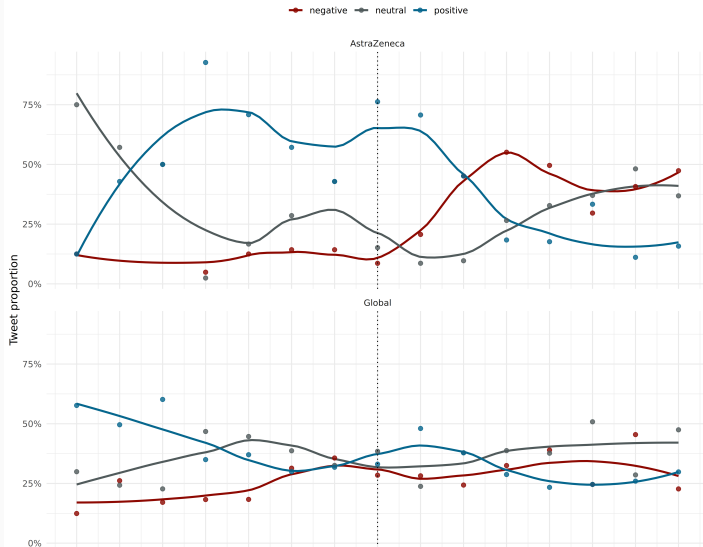
Window: -7 to +7 days | Date: November 16, 2020



RQ2 - Results: Sentiment Dynamics (AstraZeneca Suspension)

Sentiment around the announcement - AstraZeneca

Window: -7 to +7 days | Date: November 23, 2020



RQ3 - Results: Dominant Side Effects in Discourse

Discourse heavily focused on severe events. SARF analysis reveals a massive disconnect in frequency perception.

Mentioned Category	Volume	Clinical Reality (Official Guidelines)	SARF Gap
1. Thrombosis	3,280	Very rare (<1/10,000): TTS (AstraZeneca).	Extreme amplification
2. Cardiac Issues	1,603	Very rare (<1/10,000): Myocarditis (Pfizer/Moderna).	High amplification
3. Death	474	Not listed in safety profiles.	Fabrication
4. Neurological	353	Seizures not listed; facial paralysis rare.	Distortion
5. Hemorrhages	305	Not listed; menstrual issues monitored.	Uncertainty gap

RQ3 - Results: Dominant Side Effects in Discourse (Interpretation)

- These are **tweet mentions**, not clinical incidence rates.
- **Thrombosis** and **cardiac issues** dominate despite being clinically very rare - classic SARF amplification.
- **Death** appears as a salient symbol, even when absent from formal safety profiles.
- **Neurological** focus emphasizes unlisted or ambiguous symptoms, widening the perception gap.
- **Hemorrhages** illustrate uncertainty: monitored signals become salient without frequency tables.

RQ3 - Results: Brand Discourse Volume (Side Effects)

Side-Effect Attention Dynamics: The dominance of AstraZeneca and Pfizer reflects heightened public scrutiny tied to brand identity.

Social Discourse

- **AstraZeneca (7,660)** and **Pfizer (7,196)** dominated the discourse.
- **Moderna (1,970)** followed significantly behind.
- **Sanofi (1,163)** had notable volume.
- **BioNTech (256)** rarely appeared alone.

Clinical Reality (Context)

- **AstraZeneca:** Only ~**7.8M** injections.
 - *Paradox:* Highest noise for lowest usage.
- **Pfizer:** ~**123.5M** injections.
 - *Context:* High volume proportional to massive use.
- **Moderna:** ~**24.2M** injections.
- **Sanofi:** ~**6,900** injections.
 - *Note:* Discourse volume is disconnected from clinical reality.

RQ3 - Results: Brand-Specific Side Effect Associations

Distinct side-effect profiles emerged. Comparison with Official Safety Profiles (ANSM):

- **AstraZeneca**
 - *Discourse*: Strongly linked to thrombotic events.
 - *Official Profile*: **TTS** listed as “**Very Rare (<1/10,000)**”.
 - *SARF Verdict*: **Rational but Disproportionate**. The risk exists but became the sole identity of the brand despite its extreme rarity.
- **Pfizer**
 - *Discourse*: Focal point for broad concerns (cardiac, neurological, fertility).
 - *Official Profile*: **Myocarditis** is “**Very Rare**”. **Seizures** are **Absent** from the adverse reaction table.
 - *SARF Verdict*: **Distortion**. Social discourse invented frequency (making rare events seem common) and symptoms (seizures).
- **Moderna**
 - *Discourse*: Associated with cardiac issues.
 - *Official Profile*: **Myocarditis** listed as “**Very Rare (<1/10,000)**”.
 - *SARF Verdict*: **Alignment**. Discourse reflects the specific regulatory warning.

- **SARF Effects:** Risk narratives amplified online, not fully aligned with clinical data
- **Signaling Issues:** Positive announcements clashed with public risk amplification
- **Brand Reputation:** Negative events can rapidly harm brand equity, requiring transparency and engagement

Questions

Any questions?